AC444
VALUATION
AND SECURITES ANALYSIS

Week 7 Session 1
FCFE and AE Based Valuations

THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Dr Pascal Frantz
London School of Economics

HOW TO CONTACT ME

- Office: MAR.3.30
- Email Address: p.frantz@lse.ac.uk

SUGGESTED READINGS

- Chapter 7 [Palepu, Healy, and Peek];

TOPICS DISCUSSED

- The Present Value of Expected Net Dividends (PVND) Valuation Model;
- Abnormal Earnings;
- The Present Value of Expected Abnormal Earnings (PVAE) Valuation Model;
- The Present Value of Expected Abnormal NOPAT Valuation Model.

PRESENT VALUE
OF EXPECTED NET DIVIDENDS (FCFE)

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- The returns from holding a share in a company are made up of 2 components:
  - ✓ net dividends;
  - ✓ capital gains (or losses);

- Over one year, starting now at date t, the realized return r_{t+1} is thus given by:

$$r_{t+1} = (nd_{t+1} + p_{t+1} - p_t)/p_t$$

where:

- ✓ p_t denotes the share price obtaining at date t (now);
- ✓ p_{t+1} denotes the share price obtaining at date t+1 (one year from now);
- ✓ nd_{t+1} denotes the net dividend per share paid at date t=1 (one year from now);

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- Equivalently, by taking expectations, we get:

$$E(r) = [E(nd_{t+1} + p_{t+1}) - p_t]/p_t$$

- In an efficient market, the expected return of a share is equal to the cost of equity capital denoted by r_E;

- Hence, the intrinsic share price:

$$p^*_t = E[(nd_{t+1}+p_{t+1})/(1+r_E)]$$

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- In turn:

$$E[p_{t+1}] = E[(nd_{t+2}+p_{t+2})/(1+r_E)]$$

- So that:

$$p_t^* = E\left[\frac{nd_{t+1}}{1+r_E} + \frac{nd_{t+2}}{(1+r_E)^2} + \ \ldots + \frac{nd_{t+n}+p_{t+n}}{(1+r_E)^n}\right]$$

- Furthermore:

$$\lim_{n\to+\infty} E\left[\frac{p_{t+n}}{(1+r_E)^n}\right] = 0$$

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- The intrinsic price of any share now, p^*_t, can hence be computed as the present value of the stream of all future expected net dividends per share:

$$p_t^* = \sum_{i=1}^{+\infty} \frac{E(nd_{t+i})}{(1+r_E)^i}$$

with:

- ✓ nd denoting net dividends (per share), that is, cash paid in dividends and stock repurchases less cash contributions from shareholders (on a per share basis).

PRESENT VALUE OF EXPECTED NET DIVIDENDS (PVND)

- Equivalently, at any date t, the intrinsic value of a firm's equity, VE^*_t, can be expressed as the present value of all future expected net dividends ND_{t+i}:

$$VE_t^* = \sum_{i=1}^{+\infty} \frac{E(ND_{t+i})}{(1+r_E)^i}$$

NET DIVIDENDS

- Definition of Net Dividend:
  Cash Dividend
  + Cash Spent on Share Repurchases
  - Proceeds from Share Issues

- From the definition of FCFE (Free Cash Flow to Equity holders):

$$
\begin{aligned}
FCFE &= FCF + \Delta BVNFO - NFE \\
&= NOPAT - \Delta BVNOA + \Delta BVNFO - NFE \\
&= ND
\end{aligned}
$$

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- Main issues:
  - ✓ One has to forecast future expected net dividends forever;
  - ✓ Empirical evidence suggests that taking the present value of the next 10 years of net dividends only accounts for a small proportion of a typical stock's equity value;
- Main reason:
  - ✓ The firm's lifecycle.

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- Financial economists hence often take a short cut and assume that expected dividends grow at a constant rate g;

- The PVND can then be rewritten as:

$$VE_t^* = \frac{E(ND_{t+1})}{r_E - g} = \frac{(1+g)ND_t}{r_E - g}$$

- This is then referred to as the Gordon growth valuation model.

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- Proof:

  The present value of a growing perpetuity of \$1 next year growing at g is:

  $\$1/(r_E - g)$

PRESENT VALUE OF EXPECTED NET DIVIDENDS

- The latter formula is easy to use;
- It works reasonably well for stable companies that are past the high growth phase of their development.
- It however does not work for fast growing firms or for firms that pay no dividend.

APPLICATION EXERCISE 1

APPLICATION EXERCISE 1

- Biotech is an all-equity firm without any outstanding debt.
- Under its current policies, Biotech pays out all of its earnings as dividends and does not expect any growth in per-share earnings.
- Current earnings per share, eps_0, are £5.
- Its cost of equity capital, r_B, is 6.8%.
- What is the current intrinsic stock price?

APPLICATION EXERCISE 1

- It follows that:

$$p^*_0 = E(nd_1)/r_B = £5/6.8\% = £73.5$$

APPLICATION EXERCISE 1

- Biotech has a new investment opportunity which will require that 50% of its earnings are reinvested each year, but which will permanently increase its return on equity to 10%.
- If Biotech takes on this new project, what will the expected annual growth rate in earnings be?

APPLICATION EXERCISE 1

- If g denotes the growth rate in eps (sustainable growth rate) and dps, and PYR denotes the earnings payout ratio (proportion of earnings paid out as dividends), then:

$$
\begin{aligned}
g &= (1 - PYR)*ROE \\
&= 50\% * 10\% \\
&= 5\%
\end{aligned}
$$

APPLICATION EXERCISE 1 (Proof)

$ND_{t+2} = PYR*NP_{t+2}$

$ND_{t+1} = PYR*NP_{t+1}$

$NP_{t+2} = NP_{t+1} + NP_{t+1}(1- PYR)ROE$

APPLICATION EXERCISE 1 (Proof)

Hence:

$$
\begin{aligned}
g &= (ND_{t+2} - ND_{t+1})/ND_{t+1} \\
&= (NP_{t+2} - NP_{t+1})/NP_{t+1} \\
&= [NP_{t+1} + (1\text{-}PYR)NP_{t+1}ROE - NP_{t+1}]/NP_{t+1} \\
&= (1\text{-}PYR)*ROE
\end{aligned}
$$

APPLICATION EXERCISE 1

- At the moment the stock market is unaware that Biotech has this new investment opportunity.
- The Biotech board of directors however plans to announce it soon.
- When the board announces the new investment plan, what will the change in Biotech's stock price be?

APPLICATION EXERCISE 1

- From the information provided, it follows that:

$E(nd_1) = eps_0 * (1+g) * PYR$

$= £5 * (1+ 5\%) * 50\%$

$= £2.625$

- And:

$p^*_0 = E(nd_1)/[r_E - g]$

$= £2.625/(6.8\%-5\%)$

$= £145.8$

APPLICATION EXERCISE 1

- The change in stock price then is:

  Δp^*_0 = £145.8 - £73.5 = £72.3
- The proportional change in stock price furthermore is:

  $\Delta p^*_0/p^*_0$ = £72.3/£73.5 = +98.6%

APPLICATION EXERCISE 1

- This new investment plan leads to a high increase in the fundamental value of the firm intuitively because the incremental funds invested generate a return of 10% (ROE) which exceed their opportunity cost of 6.8% (r_E).

ABNORMAL EARNINGS

ABNORMAL EARNINGS

- *Abnormal earnings*, also referred to as residual earnings or abnormal or residual income, are defined as a firm's *actual earnings minus* the firm's *normal earnings (normal profit)*;

- *Normal earnings* are defined as the *cost of equity capital* (also referred to as the normal rate of return) *multiplied by the book value of equity* (obtaining at the beginning of the year);

ABNORMAL EARNINGS

$$AE_t = NP_t - (r_E * BVE_{t-1})$$

- AE_t denoting the abnormal earnings obtaining in year t;
- NP_t denoting the actual net profit (net income) obtaining in year t (net income);
- r_E denoting the cost of equity capital;
- BVE_{t-1} denoting the book value of equity obtaining at the end of year t-1 (beginning of year t);

ABNORMAL EARNINGS

$$AE_t \equiv AROE_t * BVE_{t-1} \equiv (ROE_t - r_E)\, BVE_{t-1}$$

- $AE_t = 0 \Leftrightarrow ROE_t = r_E$;
- $AE_t > 0 \Leftrightarrow ROE_t > r_E$;
- $AE_t < 0 \Leftrightarrow ROE_t < r_E$;

ABNORMAL EARNINGS

$$AE_t \equiv (ROE_t - r_E)\, BVE_{t-1}$$

- Consider a firm without any source of comparative advantage in a very competitive industry;
- This implies no supernormal profitability according to economists;
- If accounting is unbiased, it follows that $ROE_t = r_E$ and $AE_t = 0$.

ABNORMAL EARNINGS

- Abnormal earnings hence capture the net profit generated in excess of the one that would have obtained if the company did not have any source of comparative advantage, was competing in a very competitive industry, and accounting was unbiased;

ABNORMAL EARNINGS

- In a perfectly competitive industry (with unbiased accounting):
  - ✓ expected abnormal earnings are equal to zero;
  - ✓ the expected ROE is equal to the cost of equity capital r_E;
- Deviations from these levels may arise for three reasons:
  - ✓ industry conditions and a competitive strategy causing a firm to generate supernormal (subnormal) economic profits;
  - ✓ financial engineering;
  - ✓ distortions due to accounting (biased accounting).

ABNORMAL EARNINGS

- Intuitively, accounting is unbiased if it captures the underlying business reality;

- Sources of bias:

  - Rigidity in accounting rules;

  - Systematic reporting choices made by corporate managers to achieve specific objectives.

ABNORMAL EARNINGS

- If accounting is unbiased (undistorted), abnormal earnings capture the economic value added by the firm accruing to shareholders (over a period).

PRESENT VALUE
OF EXPECTED ABNORMAL EARNINGS (PVAE)

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Clean surplus accounting obtains if, at the end of any year, the book value of equity is equal to the book value of equity at the end of the previous year plus the net profit generated minus the net dividends paid during that period:

$$BVE_t = BVE_{t-1} + NP_t - ND_t$$

Statement of Equity	
BVE_{t-1}	ND_t
NP_t	BVE_t

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- If clean surplus accounting does not obtain:
  - ✓ Consider comprehensive earnings (CE_t) defined as follows:

$$CE_t = BVE_t - BVE_{t-1} + ND_t$$

Statement of Equity	
BVE_{t-1}	ND_t
CE_t	BVE_t

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- If clean surplus accounting does not obtain:
  - ✓ Then define AE_t as follows:

$$AE_t = CE_t - (r_E * BVE_{t-1})$$

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Clean surplus accounting implies that $NP_t = CE_t$;
- Clean surplus accounting does not obtain if some gains or losses, excluding transactions with shareholders, are recognized in the balance sheet statement bypassing the income statement;
- Dirty surplus accounting tends to involve transitory or one-off items;
- Dirty surplus accounting is not an issue when forecasting future statements.

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- The intrinsic value of a firm's equity can then be expressed as the sum of the current book value of equity plus the present value of expected future abnormal earnings (PVAE valuation).

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

$$VE^*_t = BVE_t + \sum_{i=1}^{i=+\infty} \frac{E_t[AE_{t+i}]}{(1+r_E)^i}$$

- with:
  - ✓ E_t denoting the expectation as of date t;
  - ✓ AE_t denoting abnormal earnings generated in year t;
  - ✓ VE_t^* denoting the intrinsic value of equity at date t (market value of equity obtaining in efficient markets);

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

Intuition

- This valuation method explicitly links a firm's fundamental equity value to:
  - ✓ the firm's future expected performance;
  - ✓ risk.

- The premium (over book value), that is, $VE_t^* - BVE_t$, is equal to the sum of discounted expected abnormal earnings;
  - ✓ the higher the level of expected abnormal earnings, the higher the premium;
  - ✓ the longer the period over which expected abnormal earnings are positive, the higher the premium.

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

Intuition

- If accounting is unbiased, investors should be willing to pay no more than the book value of equity to acquire the equity of a firm in a perfectly competitive industry;

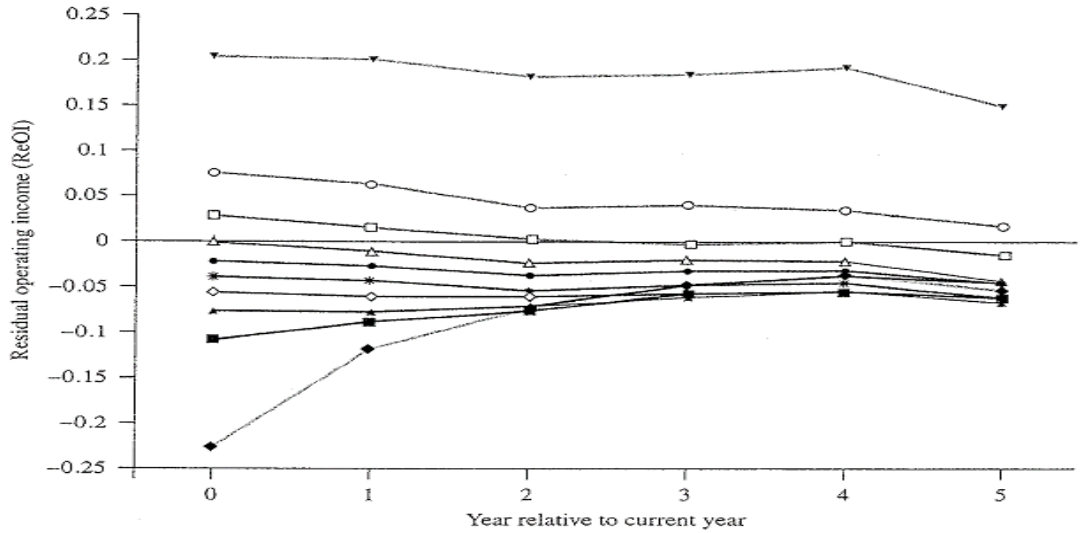
- If accounting is unbiased and the market in which the firm operates is not perfectly competitive but is expected to become so at some stage in the future, the PVAE formula can be truncated!

BEHAVIOUR OF ABNORMAL EARNINGS

FIGURE 14.2

Patterns of residual operating income (a), return on net operating assets (b), and growth in net operating assets (c) over five-year periods for NYSE and AMEX firms between 1965 and 1996.

(a) Residual operating income (ReOI). Firms with relatively high ReOI in the current year (Year 0) in the upper groups tend to maintain a high ReOI in the subsequent five years. Firms with relatively low ReOI in the current year in the lower groups tend to have low ReOI in the subsequent five years. Overall, ReOI tends to move toward zero for all groups. (ReOI is deflated by net operating assets at the beginning of Year 0.)



PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Whilst expected abnormal earnings in any year in the future depend on the set of accounting choices made, the PVAE formula generates the same valuation estimate however biased the accounting (as long as choices do not affect investors' perceptions of future prospects);

- It also generates the same intrinsic value of equity than the PVND.

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

Derivation

- Starting Point:

  The intrinsic value of a firm's equity can be expressed as the present value of all future expected net dividends.

- Then:

  Rearrange terms by expressing:

  - Net dividends as comprehensive earnings adjusted for the change in the book value of equity;
  - Comprehensive earnings as abnormal earnings adjusted for the charge for the use of equity.

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Equity research analysts often add additional assumptions to find out what the market has in mind and whether it is reasonable;

- For instance, analysts sometimes assume that expected abnormal earnings grow at a constant rate g;

- If so, the PVAE valuation formula simplifies to:

$$VE_t^* = \frac{ROE_{t+1} - g}{r_E - g} BVE_t$$

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Proof:

1. Start with:

$$VE^*_t = BVE_t + \sum_{i=1}^{i=+\infty} \frac{E_t[AE_{t+i}]}{(1+r_E)^i}$$

2. Recognise that expected abnormal earnings are growing at a rate g and hence:

$$\sum_{i=1}^{+\infty} \frac{AE_{t+i}}{(1+r_E)^i} = \frac{AE_{t+1}}{r_E - g} = \frac{(ROE_{t+1} - r_E)BVE_t}{r_E - g}$$

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Proof (Continued)

3. It follows that:

$$VE*_t = \left[1 + \frac{ROE_{t+1} - r_E}{r_E - g}\right] BVE_t$$

and:

$$VE_t^* = \frac{ROE_{t+1} - g}{r_E - g} BVE_t$$

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Consider Indigo Inc;

- Its current market price is $15 per share and its book value is $5 per share.

- The consensus amongst analysts is that:
  - ✓ the book value will grow by 10% per year indefinitely;
  - ✓ the cost of equity is 15%;

- There are however differences of opinion about future expected ROE (which are expected to remain at the same level indefinitely).

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Consider Jo Blogg, an analyst with CSFB;

- Jo may hence use the previous formula (replacing the fundamental value with the market value) to find out which future expected ROE is priced in the capital market:

$$ROE = r_E + \left(\frac{VE_t}{BVE_t} - 1\right)(r_E - g)$$

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Hence:

  ROE = 15% + (3 – 1)(15% - 10%) = 25%

- Jo may now make an investment recommendation:
  - ✓ "Sell" if his estimate is 20%;
  - ✓ "Buy" if his estimate is 30%.

PRESENT VALUE OF EXPECTED ABNORMAL EARNINGS

- Intuition:

  - ✓ Intrinsic value per share of Indigo if ROE = 20%:

    $ve^*_t = [(20\% - 10\%)/(15\% - 10\%)]\$5 = \$10$

  - ✓ Intrinsic value of Indigo per share if ROE = 30%:

    $ve^*_t = [(30\% - 10\%)/(15\% - 10\%)]\$5 = \$20$

APPLICATION EXERCISE 2

APPLICATION EXERCISE 2
R&D Inc.

- Consider R & D Inc, a biotech start up. This firm:
  - ✓ Incurs expenditure in R&D of $50m in the first year of activity;
  - ✓ Has an opening book value of equity of $1bn;
  - ✓ Generates income (before R & D expenses) of $205m in year 1 and $221m in year 2, at the end of which it is liquidated;
  - ✓ Has a cost of equity capital of 10%;
  - ✓ Pays $155m in cash dividends at the end of the first year.
- Assume that clean surplus accounting obtains.

APPLICATION EXERCISE 2
R&D Inc.

- Consider the following accounting policies for research and development:

  - ✓ Option A:
    The firm expenses R & D in the year in which it is incurred;

  - ✓ Option B:
    The firm capitalises R & D and amortizes $25m in each year;

APPLICATION EXERCISE 2
R&D Inc.

- Construct statements of equity for R&D Inc for both options.

Option A: Expensing R & D as Incurred

Year 1

BVE_0 $1,000m	BVE_1 $1,000m
NP_1 $155m	ND_1 $155m

Year 2

BVE_1 $1,000m	ND_2 $1,221m
NP_2 $221m	

$$BVE_1 = BVE_0 + NP_1 - ND_1$$
$$ND_2 = BVE_1 + NP_2$$

Option B: Capitalizing and Amortizing R & D

Year 1

BVE_0 $1,000m	BVE_1 $1,025m
NP_1 $180m	ND_1 $155m

Year 2

BVE_1 $1,025m	ND_2 $1,221m
NP_2 $196m	

$$BVE_1 = BVE_0 + NP_1 - ND_1$$
$$ND_2 = BVE_1 + NP_2$$

APPLICATION EXERCISE 2
R&D Inc.

- Calculate ROE and AE generated by R&D Inc. in each year of activity using both options.

Option A: Expensing R & D as Incurred

Year 1

BVE_0 $1,000m	BVE_1 $1,000m
NP_1 $155m	ND_1 $155m

Year 2

BVE_1 $1,000m	ND_2 $1,221m
NP_2 $221m	

$ROE_1 = NP_1/BVE_0 = \$155m/\$1,000m = 15.5\%$

$ROE_2 = NP_2/BVE_1 = \$221m/\$1,000m = 22.1\%$

$AE_1 = NP_1 - r_E{*}BVE_0 = \$155m - \$100m = \$55m$

$AE_2 = NP_2 - r_E{*}BVE_1 = \$221m - \$100m = \$121m$

Option B: Capitalizing and Amortizing R & D

Year 1

BVE_0 $1,000m	BVE_1 $1025m
NP_1 $180m	ND_1 $155m

Year 2

BVE_1 $1,025m	ND_2 $1,221m
NP_2 $196m	

$ROE_1 = NP_1/BVE_0 = \$180m/\$1,000m = 18.0\%$
$ROE_2 = NP_2/BVE_1 = \$196m/\$1,025m = 19.1\%$
$AE_1 = NP_1 – r_E*BVE_0 = \$180m – \$100m = \$80m$
$AE_2 = NP_2 – r_E*BVE_1 = \$196m – \$102.5m = \$93.5m$

APPLICATION EXERCISE 2
Performance Evaluation

ROE (BY)	YEAR 1	YEAR 2
Option A	15.5%	22.1%
Option B	18.0%	19.1%

AE	YEAR 1	YEAR 2
Option A	$55m	$121m
Option B	$80m	$93.5m

APPLICATION EXERCISE 2
Performance Evaluation

- Performance in both year 1 and year 2, whether measured by ROE or AE, depends on choice of accounting policy for R & D.

APPLICATION EXERCISE 2
Performance Evaluation

- Show that the PVAE formula provides the same outcome regardless of whether R & D Inc expenses R & D expenditure as incurred or capitalizes and amortizes R & D expenditure.

Option A: Expensing R & D as Incurred

Year 1

BVE_0 $1,000m	BVE_1 $1,000m
NP_1 $155m	ND_1 $155m

Year 2

BVE_1 $1,000m	ND_2 $1,221m
NP_2 $221m	

$$V_E^* = \$1000m + \frac{\$55m}{1.1} + \frac{\$121m}{1.1^2} = \$1150m$$

Option B: Capitalizing and Amortizing R & D

Year 1

BVE_0 $1000m	BVE_1 $1025m
NP_1 $180m	ND_1 $155m

Year 2

BVE_1 $1025m	ND_2 $1221m
NP_2 $196m	

$$V_E^* = \$1000m + \frac{\$80m}{1.1} + \frac{\$93.5m}{1.1^2} = \$1150m$$

APPLICATION EXERCISE 2
R&D Inc.

$$V_t^E = \frac{ND_1}{1+r_E} + \frac{ND_2}{(1+r_E)^2} = \frac{\$155m}{1.1} + \frac{\$1221m}{1.1^2} = \$1150m$$

- The valuation obtained from the PVAE formula is hence exactly the same as the one obtained using the PVED formula.

ABNORMAL NOPAT

ABNORMAL NOPAT
Stylised Income statements

Stylised Income Statement	
Net Operating Profit After Tax (NOPAT)	(Net Financial Expense) NFE
	(Comprehensive Earnings) CE

ABNORMAL NOPAT
Stylised Balance Sheets

Stylised Balance Sheet	
Operating Assets (OA)	Operating Liabilities (OL)
Financial Assets (FA)	Financial Liabilities (FL)
Total Assets (TA = OA+FA)	Total Liabilities (TL = OL+FL)

Reformulated Balance Sheet	
Net Operating Assets (NOA = OA-OL)	Net Financial Obligations (NFO = FL-FA)
	Equity

ABNORMAL NOPAT
Reformulated Balance Sheets

Reformulated Balance Sheet (Book Values)	
BVNOA	BVNFO
	BVE

Reformulated Balance Sheet (Intrinsic Values)	
EV	Intrinsic Value of NFO
	VE

ABNORMAL NOPAT

- *Abnormal NOPAT,* ANOPAT, (also referred to as Abnormal Operating Income, AOI, or Abnormal Operating Earnings) is defined as a firm's *Actual Net Operating Profit After Tax (Operating Income) minus* the firm's *normal NOPAT*;

- *Normal NOPAT* is defined as the *firm's cost of capital* (also referred to as the firm's cost of capital or normal rate of return) *multiplied by the book value of the firm's net operating assets* (obtaining at the beginning of the year);

ABNORMAL NOPAT

$$ANOPAT_t = NOPAT_t - (r * BVNOA_{t-1}) = (RNOA_t - r)\ BVNOA_{t-1}$$

- $ANOPAT_t$ denoting the abnormal net operating profit after tax (AOI after tax) obtaining in year t;
- $NOPAT_t$ denoting the actual net operating profit after tax (OI after tax) obtaining in year t;
- r denoting the firm's cost of capital which can be estimated as the weighted average cost of capital of all financial claims;
- $BVNOA_{t-1}$ denoting the book value of the firm's net operating assets obtaining at the end of year t-1 (beginning of year t).

ABNORMAL NOPAT

- Abnormal NOPAT hence capture the operating profit (NOPAT) generated in excess of the one that would have obtained if the company did not have any source of comparative advantage, was competing in a very competitive industry, and accounting was unbiased;

ABNORMAL NOPAT

- When Abnormal NOPAT is derived under Unbiased Accounting, it is also referred to as Economic Value Added (EVA)!

PRESENT VALUE OF EXPECTED ABNORMAL NOPAT (PVANOPAT)

PRESENT VALUE OF EXPECTED ABNORMAL NOPAT

$$EV_t^* = \text{BVNOA}_t + \sum_{i=1}^{+\infty} \frac{E_t(ANOPAT_{t+i})}{(1+r)^i}$$

with:

- ✓ EV_t^* denoting the enterprise value at date t (intrinsic value of the firm's net operating assets);
- ✓ $BVNOA_t$ denoting the book value of the firm's net operating assets as of date t;
- ✓ E_t denoting the expectation as of date t;
- ✓ $ANOPAT_t$ denoting the abnormal NOPAT generated by the firm's net operating assets in year t-1;
- ✓ r denoting the firm's cost of capital.

PRESENT VALUE OF EXPECTED ABNORMAL NOPAT

- From the enterprise value, one obtains an intrinsic value of equity by:
  - ✓substracting the value of financial obligations such as debt;
  - ✓adding the value of financial assets held by the firm.

PRESENT VALUE OF EXPECTED ABNORMAL NOPAT

- When accounting is unbiased, the Present Value of expected Abnormal NOPAT formula is also referred to as the Present value of expected Economic Value Added (EVA) formula!